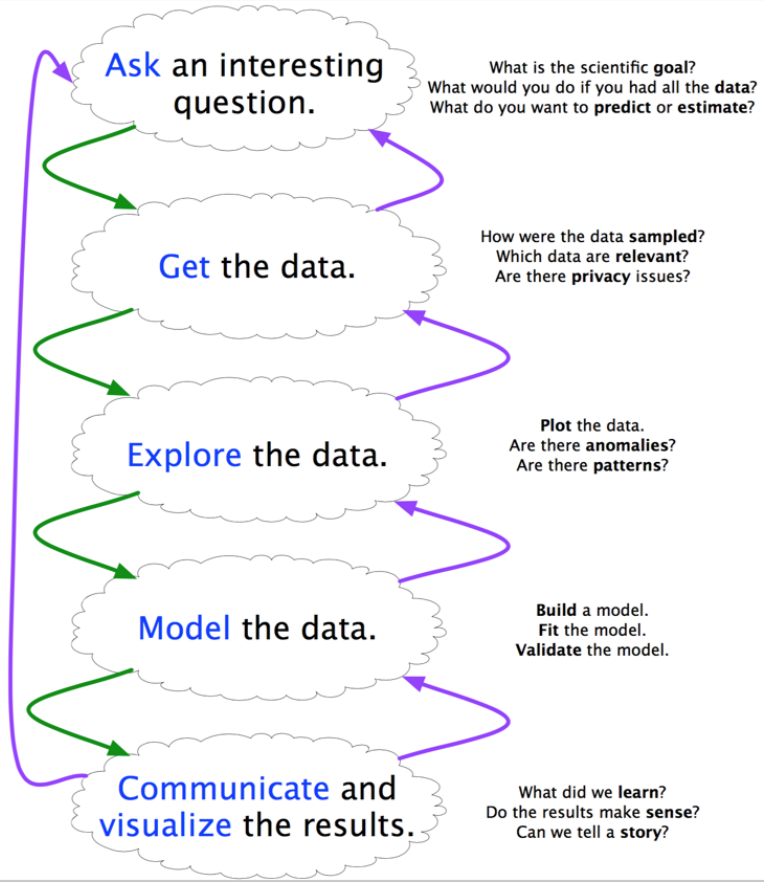
UNIVERSITY OF CYPRUS
DEPARTMENT OF COMPUTER SCIENCE

EPL 445 FINAL PROJECT - 2025

i. Extract from the contract

II. Final Project (25%, proposal and presentations 5%, report and system demo 20%): Students will work individually or in groups on their project to be presented near the end of the semester. A final technical paper (20% of the final grade) for the project should be submitted, as well as a detailed presentation/demo of the system. Students will submit a project proposal and present their project progress to the class (5% of the final grade). Students can select their own project topic or choose from a list of projects suggested by the instructor. Projects should focus on the design and implementation of systems based on image and/or video processing and analysis that will include intelligent analysis to facilitate decision making.



ii. Topics

The following topics can be investigated in different areas (e.g. sports – football, sports – road racing, sports – racing) (face recognition – pain, face recognition – feelings, face recognition – behaviour patterns):

1. Mobile video processing and analysis.
2. Face Recognition in videos
3. Object Recognition in video
4. Semantic Segmentation
5. Compare and evaluate video codecs

6. Identifying main scenes in videos (video summarization)
7. Search for images in digital video libraries
Content Based Image Retrieval (Searching in Digital Libraries)
8. Background – Foreground Extraction in video
9. Automatic car driving – auto pilot
10. Other Subjects

Each group should choose a different topic.

iii. Datasets

You need to find video datasets to develop your system and evaluate its performance.

ChatGPT recommendation – Note: not verified:

For university senior year projects in video processing and analysis, here are some open source video datasets:

- **UCF101:** A popular dataset for action recognition featuring 101 action categories from YouTube videos.
- **HMDB51:** Contains a diverse range of human motion videos, useful for activity recognition tasks.
- **Kinetics-400:** A large-scale dataset with 400 human action classes curated from YouTube, ideal for deep learning projects.
- **ActivityNet:** Provides a comprehensive benchmark for video classification and temporal action detection.
- **YouTube-8M:** A large, labeled dataset of YouTube videos suitable for video classification and representation learning.
- **DAVIS:** Focused on video object segmentation, useful for projects involving object tracking and segmentation.
- **VOT (Visual Object Tracking):** A dataset used in the Visual Object Tracking challenges, beneficial for tracking algorithm development.
- **AVA:** Offers spatio-temporal annotations for human actions, ideal for advanced action detection research.

  These datasets are widely used in the research community, and they offer a variety of challenges and opportunities for hands-on projects in video processing and analysis.

End of ChatGPT recommendation.

iv. Deep Learning Algorithms

ChatGPT recommendation – Note: not verified:

Several open source deep learning algorithms are available to process video datasets, including:

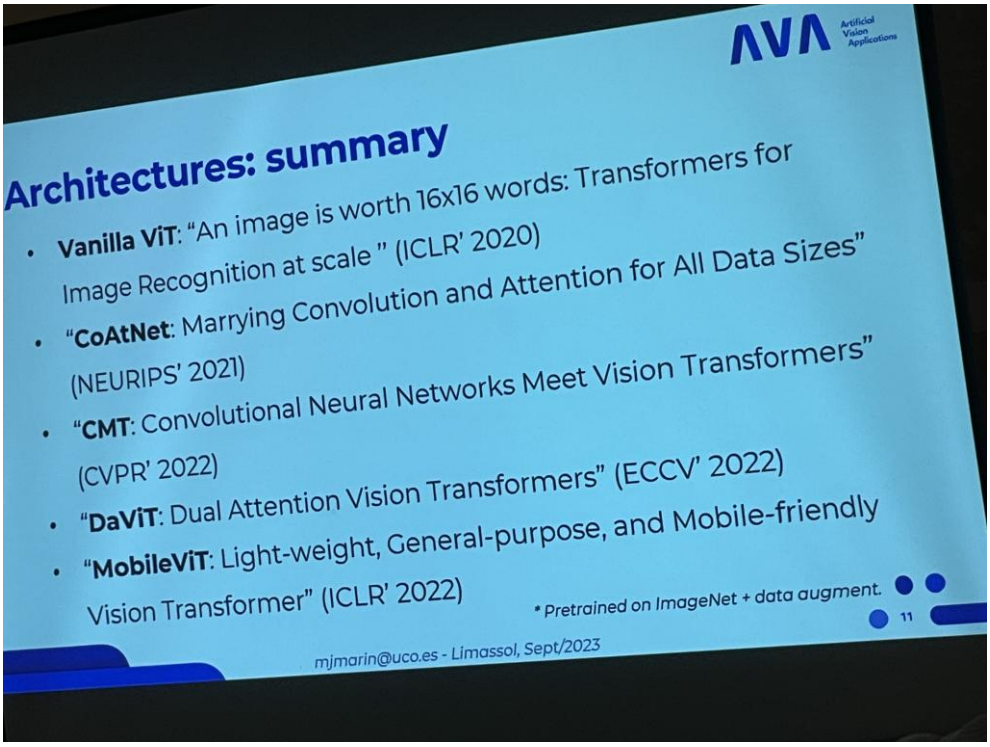
- **C3D (Convolutional 3D Networks):** For extracting spatio-temporal features from video clips.

- **I3D (Inflated 3D ConvNets):** Which adapts 2D CNN filters to 3D, available in both TensorFlow and PyTorch.
- **Two-Stream Networks:** That simultaneously process RGB frames and optical flow to capture spatial and temporal information.
- **Temporal Segment Networks (TSN):** Designed to capture long-range temporal structures in videos.
- **SlowFast Networks:** A framework from Facebook AI Research that captures both slow and fast features in video data.
- **R(2+1)D Networks:** Which factorize 3D convolutions into 2D spatial and 1D temporal convolutions.
- **3D ResNets:** Extending the popular 2D ResNet architecture to handle video data.

These algorithms, many of which are integrated into open source libraries like PyTorchVideo and Torchvision, provide robust frameworks for tackling tasks such as action recognition, segmentation, and classification in video processing projects.

End of ChatGPT recommendation.

You can try also: Transformers in image and video processing – Examples of transformer architectures in image and video processing – there are many more:



v. Report and Work Structure [20%]

0. Abstract

1. Introduction (2%)

The introduction will include a brief bibliographic reference that will include the major and most recent applications in the respective domain.

2. Methodology (6%)

You will theoretically develop the methodology-algorithms you apply + coding of the algorithms in Python. All applications should be able to handle video and should be web-based.

3. Results (6%)

You will present the results of your methodology.

4. Discussion (3%)

Develop the reasons you chose this methodology (pros and cons) and comment on your results.

5. Bibliography

6. Presentation Report (3%)

- A description of your project (**4 pages,** two columns word document), specifying in detail your goals, approach, milestones, evaluation methodology and experimental results. A document that describes how you did your experiments and what are your obtained results.
- Paper format, to be based on IEEE conference paper template.
  (see https://www.ieee.org/conferences/publishing/templates.html,
  select Microsoft Word - A4 (DOC, 30 KB) Updated Jan 2019)

vi. Presentations

Interim Presentations no. 1: 28-3-2025 [2.5% out of 5%]; 3-5 minutes (introduction, methodology, resources required to run your code: i.e. (open-source code is available in Python, needs GPU to be run (give estimated timings if available); (b) trained models are available (give estimated timings if available)).

Interim Presentation no. 2 – new date: 4 or 25-4-2025 [2.5% out of 5%]; 3-5 minutes – FIRST RUNS – TRY TO RUN ONLY THE LIGHT VERSIONS OF YOUR CODE. This task will allow you to verify that your code is working properly. Study the PERFORMANCE METRICS (i.e. Accuracy, Sensitivity, Specificity, ROC curves, AUC, F-Score, …). THESE METRICS NEED TO BE GIVEN FOR THE EVALUATION SET OR TESTING SET and possibly for the training set. You can also try to run the PRE-TRAINED MODELS if it takes a lot of time to train your models.

OPTIONAL – You do not have to do it for the 27-11-2023 presentation.

Final Project Presentation – to be announced – end of May: Delivery of Work [20%]

Final presentation 10-15 minutes along with the demo of the application.

vii. Final Examination

Question related to the topic of final project in the final examination.

viii. Groups

To be completed.